Surprisingly, the CMRR of the 741 and 107 is further degraded by load element imbalance. High sensitivity to load imbalances is an indication that the amplifier bias current is changing with common-mode voltage [7] (i.e., high common-mode transconductance).

To understand the source of high load sensitivity in the 741 and 107 stages, the sensitivities of amplifier bias current to variation in collector-emitter potential were computed and are presented in Table III. In the 741 amplifier, the quiescent current changes are due to the finite Early voltage of the p-n-p current mirror device. In the 107 stage, the nonzero output conductance of the p-n-p output devices causes the current change. These bias current changes also increase the sensitivity of CMRR to parameter imbalances in the input and output devices of the amplifiers. In the pair and cascode stages, the bias current is fixed by the high-impedance current source. Bias current variations induced by the active devices of these amplifiers are reduced by factors of β or β^2 relative to the 107 and 741 stages and the sensitivities to load imbalances are negligible.

IV. A Design Example

The 741 input stage was designed to provide the maximum protection of the amplifier from destructive input conditions, with as little degradation of other performance as possible. Improved processing allowed the design of the 741 to be used and still maintain reasonable offset voltage specifications. However, as demonstrated in Section III, the gain and CMRR of the 741 were degraded. The analysis tool previously discussed has been used to suggest and then test improvements to this stage.

The sensitivities of Tables I and II show that one area in which the 741 performance could be improved is CMRR. As previously discussed, the p-n-p current mirror contributes strongly to CMRR through the Early effect. This suggests an improvement can be obtained by replacing the p-n-p current mirror by either a cascode source or a Wilson source [8] as in Fig. 4. An analysis of this idea is presented in Table IV. For a given set of parameter imbalances, the CMRR can be improved by almost 10 dB. Sensitivity to load imbalance has been eliminated. Also, CMRR is now limited by the Early effect in both the n-p-n input pair and p-n-p output pair. This suggests replacing the n-p-n and p-n-p pairs with cascode configurations. Further, the high output resistance of the cascode stage can be utilized to improve the gain of the stage. This may be achieved by replacing the active load with a Wilson load.

The results of an analysis of the improved stage of Fig. 4 is given in Table V. The stage should achieve a very high CMRR, a gain of more than 25 000, and offset characteristics similar to the 741 amplifier. Indeed, a stage was breadboarded using RCA integrated devices, and achieved a gain $>40\,000$, CMRR >140 dB and $V_{0s}<1$ mV.

V. Summary and Conclusions

The differential amplifier characterization program allows one to effectively test and evaluate design ideas. Many ideas can be tested in a short time to arrive at a design which appears to meet the designers goals. Shortcomings of various designs are readily apparant during the initial design process, and the availability of the sensitivity information helps elucidate problem areas and develop possible solutions.

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Harmonic Output of Class-*E* RF Power Amplifiers and Load Coupling Network Design

NATHAN O. SOKAL AND FREDERICK H. RAAB

Abstract—**Class-*E* RF power amplifiers have advantages of explicit *a priori* designability, reproducibility, high efficiency, and low-stress operation of the power output devices. A previous publication [1] gave load network design equations, but only minimal information on harmonic output. We give here information on harmonic output and explicit design criteria for harmonic-suppression networks to be used with class-*E* power amplifiers.**

I. Introduction

Class-*E* RF power amplifiers [1], [2] have the advantages of 1) explicit *a priori* designability, reproducibility, and low sensitivity to component tolerances (particularly transistor parameters), leading to low manufacturing cost; 2) low stress on the power output transistors, leading to high reliability and consequent low cost of equipment warranty and maintenance; and 3) high efficiency, leading to small size and light weight of batteries and heat sinks in portable equipment, and to low cost of power consumption for high-power fixed-station equipment.

A previous publication [1] gave load network design equations, but only minimal information on harmonic output. Quantitative information on harmonic output of the class-*E* circuit is needed in order to be able to provide explicit *a priori* designs which will meet specified harmonic output requirements. That information is given here, together with explicit design criteria for harmonic suppression networks.

Manuscript recieved May 17, 1976; revised September 18, 1976.
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TABLE I
HARMONIC COMPONENTS OF COLLECTOR VOLTAGE AND OF INPUT CURRENT TO LOAD COUPLING NETWORK

n	c_n	c_n/c_1	I_n/I_1 with Q_L=							
			1	2	3	4	5	7.5	10	3 to ∞
1	1.639	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2	0.8477	0.5172	6.02 E-01	2.70 E-01	1.74 E-01	1.28 E-01	1.02 E-01	6.69 E-02	4.99 E-02	≈0.51/Q_L
3	0.2222	0.1356	1.39 E-01	4.76 E-02	2.87 E-02	2.05 E-02	1.60 E-02	1.03 E-02	7.58 E-03	≈0.080/Q_L
4	0.1432	0.08737	7.61 E-02	2.31 E-02	1.36 E-02	9.63 E-03	7.46 E-03	4.77 E-03	3.51 E-03	≈0.037/Q_L
5	0.07997	0.04879	3.61 E-02	1.03 E-02	6.01 E-03	4.24 E-03	3.28 E-03	2.09 E-03	1.54 E-03	≈0.016/Q_L
6	0.05907	0.03604	2.30 E-02	6.35 E-03	3.68 E-03	2.59 E-03	2.00 E-03	1.27 E-03	9.35 E-04	≈0.010/Q_L
7	0.04079	0.02489	1.39 E-02	3.76 E-03	2.17 E-03	1.53 E-03	1.18 E-03	7.50 E-04	5.50 E-04	≈0.0059/Q_L
8	0.03235	0.01974	9.81 E-03	2.61 E-03	1.51 E-03	1.06 E-03	8.15 E-04	5.18 E-04	3.80 E-04	≈0.0041/Q_L
9	0.02467	0.01505	6.72 E-03	1.77 E-03	1.02 E-03	7.16 E-04	5.52 E-04	3.50 E-04	2.57 E-04	≈0.0028/Q_L
10	0.02045	0.01248	5.05 E-03	1.32 E-03	7.60 E-04	5.33 E-04	4.11 E-04	2.61 E-04	1.91 E-04	≈0.0021/Q_L

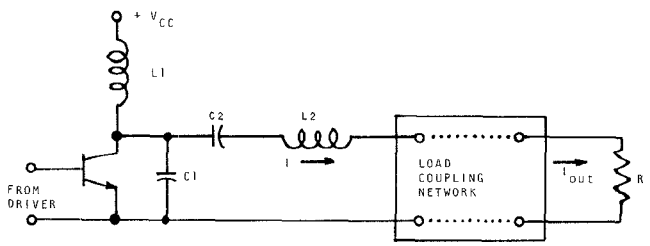


Fig. 1. Elementary version of class-E RF power amplifier.

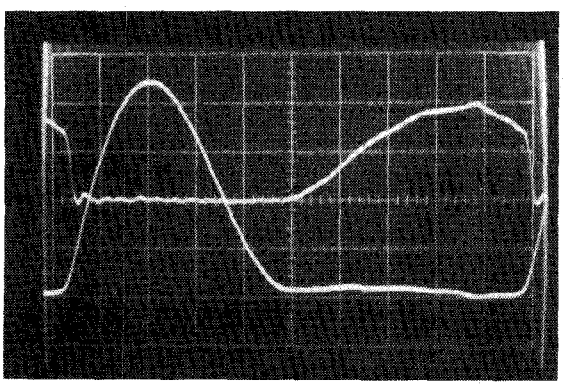


Fig. 2. V_{CE} (lower trace) and I_C (upper trace) waveforms for circuit of Fig. 1 operating at 10.5 MHz. Vertical: 10 V or 0.5 A/div; horizontal: 10 ns/div.

II. HARMONIC SPECTRUM OF COLLECTOR VOLTAGE

The published elementary version of the class-E amplifier is shown in Fig. 1. Fig. 2 shows an oscilloscope photograph of the collector voltage waveform. Table I shows the computed spectrum of the collector voltage waveform (taken from [3]), where c_n is the ratio of the nth harmonic voltage to the dc voltage V_{CC}. (The waveform shape depends slightly on $Q_L \equiv X_{L2}/R$; the spectrum calculations were performed on the waveform shape which exists in the asymptotic case of Q_L approaching infinity. Thus the subsequent equations are accurate through first order in the quantity $1/Q_L$.)

III. HARMONIC SPECTRUM OF INPUT CURRENT TO LOAD COUPLING NETWORK

The input current to the load coupling network is the current I in the $C2 - L2$ branch of the load network. At harmonic frequencies, the input impedance of the load coupling network is typically much smaller than the impedance of the $L2 - C2$ branch (hereafter "Z"), e.g., by a factor of ten or more. Hence, a good approximation to the harmonic content of the current in the $C2 - L2$ branch, without yet having *a priori* knowledge of the input impedance of the load coupling network, is

$$I_n/I_1 \approx (c_n/c_1)\,(Z_1/Z_n) \tag{1}$$

where subscripts n and 1 refer to the nth harmonic and the fundamental, respectively. c_n/c_1 values are given in Table I. Z_1/Z_n is obtained from [1, eq. (2) and (4)] by a straightforward derivation as

$$Z_1/Z_n \approx \left(\frac{1.42}{nQ_L}\right)\left[\frac{1}{(1 - 1/n^2) - (0.66 - 2.08/n^2)/Q_L}\right]. \tag{2}$$

The resulting I_n/I_1 is tabulated in Table I for several representative values of Q_L, using c_n/c_1 from Table I and Z_1/Z_n from (2). The last column of Table I gives a shortcut approximation (±10 percent) to the Table I values.

For values of n and/or Q_L larger than 10 (not in Table I), I_n/I_1 can be extrapolated beyond the range of the table by using a few simple rules of proportionality. First, for $n > 10$: because the voltage waveform contains no discontinuities, the harmonic amplitudes vary approximately as $1/n^2$; hence $c_n \approx (10/n)^2 c_{10}$. Z_n varies approximately as n; hence $1/Z_n \approx (10/n)\,(1/Z_{10})$. Consequently,

$$I_n/I_1 \approx (10/n)^3 (I_{10}/I_1). \tag{3}$$

Next, for $Q_L > 10$: because the reactance of $L2 - C2$ is proportional to Q_L, the harmonic current varies as $1/Q_L$. Hence

$$(I_n/I_1)_{Q_L} \approx (10/Q_L)\,(I_n/I_1)_{Q_L=10}. \tag{4}$$

Finally, if both n and Q_L are larger than 10,

$$(I_n/I_1) \approx (10/n)^3 (10/Q_L)\,(I_{10}/I_1)_{Q_L=10}. \tag{5}$$

The error in the approximation (1) is conservative, i.e., the actual harmonic current into the coupling network will be slightly less than that given by (1) and Table I.

IV. CURRENT TRANSFER RATIO REQUIRED OF LOAD COUPLING NETWORK

Knowing 1) the harmonic spectrum of the current entering the input port of the load coupling network (from I_n/I_1 in Table I) and 2) the allowable harmonic output to the load (from the specified performance requirements), one obtains the current transfer ratio required of the load coupling network (as terminated by the specified load) as

$$\frac{(I_{out}/I)_n}{(I_{out}/I)_1} \leq \frac{1}{S_n(I_n/I_1)}, \tag{6}$$

where S_n is the required minimum ratio of fundamental output current to nth harmonic output current. For example, if the nth harmonic output is to be at least 60 dB below the

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TABLE II
EXAMPLE, CURRENT TRANSFER RATIO REQUIRED FOR S_n = 1000 (60 dB) OR 100 (40 dB) AND Q_L = 5

n	MAXIMUM VALUE OF $\frac{(I_{out}/I)_n}{(I_{out}/I)_1}$	
	S_n = 1000 (60 dB)	S_n = 100 (40 dB)
2	0.00980	0.0980
3	0.0625	0.625
4	0.134	1.34
5	0.305	3.05
6	0.500	5.00
7	0.847	8.47
8	1.23	12.3
9	1.81	18.1
10	2.43	24.3

carrier, S_n = 1000. Now, given a chosen coupling network topology, the required element values can be found by imposing (6) as a design requirement, plus other requirements such as output/input impedance transformation ratio, low-pass cutoff frequency, variation of fundamental current transmission ratio versus frequency within a specified frequency band, and load presented to the power amplifier when the output port of the load coupling network is grossly misterminated (e.g., by a shorted or open transmission line to the antenna). Standard network analysis procedures are applicable here.

Once the network has been designed by this approximate method, the approximations of (1) and Table I can be refined (if desired) by accounting for 1) the nonzero input impedance of the coupling network at harmonic frequencies, and 2) the effects of modifying the values of $C2$ and/or $L2$ to compensate for possible capacitive and/or inductive input impedance of the coupling network within the operating frequency band (see [1, paragraph after (4)]).

V. EXAMPLE

We give here an example of the use of (6) and the data in Table I to specify the required current transfer ratio of the load coupling network. For this example, we assume the following set of requirements:

1) all harmonic components of the current into R_L must be at least 60 dB (first case) or 40 dB (second case) below the fundamental, and

2) Q_L = 5 has been chosen for the load network design.

Requirement 1) results in S_n = 1000 (first case) or 100 (second case) for all $n \geqslant 2$. Table II shows the needed sets of values of $(I_{out}/I)_n/(I_{out}/I)_1$, obtained from (6) and the data in the Q_L = 5 column of Table I.

The authors intend to publish subsequently design examples with comparisons of calculated and measured performance.

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Comments on "Rapid Determination of Emitter- and Collector-Bulk Resistances"

S. ASHOK

In the above correspondence[1], a simple curve tracer technique is presented to rapidly determine the emitter- and collector-bulk resistances of a bipolar transistor. Briefly, the emitter resistance R_{SE} is determined from the voltage intercept of V_{CE} - I_C characteristics under heavy base overdrive, while the sum of collector and emitter resistances is found from the slope at low levels of I_C.

The topological symmetry of the transistor suggests an alternate approach, viz., to simply interchange the collector and emitter terminals. Now the collector resistance R_{SC} is found from the voltage intercept, independent of the emitter resistance. This method is particularly advantageous when the collector resistance is appreciably smaller than the emitter resistance. Tests on a number of small-signal bipolar transistors indicate that almost invariably the value of R_{SC} obtained from the method described in the above-mentioned correspondence[1] depends strongly on the base overdrive chosen. On the other hand, R_{SC} obtained from V_{EC} intercept is relatively independent of the base overdrive level. Admittedly, in the inverted mode of operation the collector acts as an emitter and the path for current flow can differ significantly from that for the normal mode depending on the transistor structure. Nevertheless, the constant value of R_{SC} obtained from the inverted mode data gives a measure of the collector-bulk resistance, and may possibly be correlated with other measurements.

The measurement of open-circuit V_{CE} and V_{EC} to obtain the emitter- and collector-bulk resistances, respectively, has been reported before [1], [2]. This technique is also amenable to the determination of source and drain parasitic resistances of JFET's.

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Author's Reply[2]

J. S. T. HUANG

We thank Dr. Ashok for bringing to our attention an alternative method for measuring R_{SC}. It may be true that the "slope" method presented in our correspondence[1] could become unreliable when the collector resistance is appreciably smaller than the emitter resistance. However, IC transistors with such characteristics are exceptions rather than rules. When a transistor is operated in the inverted mode, the current is injected over a much larger area, i.e., the normal base-collector junction. The current flow path can result in the lowering of the equivalent R_{SC} obtained from the V_{EC} intercept. We measured R_{SC} in both ways. The result is 9.7 Ω for R_{SC} by the inverted transistor technique as compared to 11.6 Ω obtained from the "slope" technique.

[2]Manuscript received July 26, 1976.

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